

1. In a random effects model, define the composite error $u_{it} = a_i + v_{it}$, where a_i is uncorrelated with v_{it} and the v_{it} have constant variance σ_v^2 and are serially uncorrelated. Define $e_{it} = u_{it} - \lambda \bar{u}_i$, where λ is

$$\lambda = 1 - \left[\frac{\sigma_v^2}{\sigma_v^2 + T\sigma_a^2} \right]^{1/2}$$

Show that the e_{it} are serially uncorrelated.

(i) $E(e_{it}) = E(u_{it} - \lambda \bar{u}_i) = E(u_{it}) - \lambda E(\bar{u}_i) = 0$ because $E(u_{it}) = 0$ for all t by the (mild) assumption.

(ii) $\text{Var}(u_{it} - \lambda \bar{u}_i) = \text{Var}(u_{it}) + \lambda^2 \text{Var}(\bar{u}_i) - 2\lambda \cdot \text{Cov}(u_{it}, \bar{u}_i) = \sigma_u^2 + \lambda^2 E(\bar{u}_i^2) - 2\lambda E(u_{it} \bar{u}_i)$.
Now, $\sigma_u^2 = E(u_{it}^2) = \sigma_a^2 + \sigma_v^2$ and

$$E(u_{it} \bar{u}_i) = T^{-1} \sum_{s=1}^T E(u_{it} u_{is}) = \sigma_a^2 + \sigma_v^2 / T.$$

Therefore, $E(\bar{u}_i^2) = T^{-1} \sum_{s=1}^T E(u_{it} \bar{u}_i) = \sigma_a^2 + \sigma_v^2 / T$. Now, we can collect terms:

$$\text{Var}(u_{it} - \lambda \bar{u}_i) = (\sigma_a^2 + \sigma_v^2) + \lambda^2 (\sigma_a^2 + \sigma_v^2 / T) - 2\lambda (\sigma_a^2 + \sigma_v^2 / T)$$

Now, it is convenient to write $\lambda = 1 - \sqrt{\eta} / \sqrt{\gamma}$, where $\eta \equiv \sigma_v^2 / T$ and $\gamma \equiv \sigma_a^2 + \sigma_v^2 / T$. Then

$$\begin{aligned} \text{Var}(u_{it} - \lambda \bar{u}_i) &= (\sigma_a^2 + \sigma_v^2) - 2\lambda (\sigma_a^2 + \sigma_v^2 / T) + \lambda^2 (\sigma_a^2 + \sigma_v^2 / T) \\ &= (\sigma_a^2 + \sigma_v^2) - 2(1 - \sqrt{\eta} / \sqrt{\gamma}) \gamma + (1 - \sqrt{\eta} / \sqrt{\gamma})^2 \gamma = (\sigma_a^2 + \sigma_v^2) + \eta - \gamma = \sigma_v^2 \end{aligned}$$

This is what we wanted to show.

(iii) We must show that $E(e_{it} e_{is}) = 0$ for $t \neq s$.

$$\begin{aligned} \text{Now } E(e_{it} e_{is}) &= E[(u_{it} - \lambda \bar{u}_i)(u_{is} - \lambda \bar{u}_i)] = E(u_{it} u_{is}) - \lambda E(\bar{u}_i u_{is}) - \lambda E(u_{it} \bar{u}_i) + \lambda^2 E(\bar{u}_i^2) = \sigma_a^2 - \\ &2\lambda (\sigma_a^2 + \sigma_v^2 / T) + \lambda^2 E(\bar{u}_i^2) = \sigma_a^2 - 2\lambda (\sigma_a^2 + \sigma_v^2 / T) + \lambda^2 (\sigma_a^2 + \sigma_v^2 / T). \end{aligned}$$

The rest of the proof is very similar to part (ii):

$$\begin{aligned} E(e_{it} e_{is}) &= \sigma_a^2 - 2\lambda (\sigma_a^2 + \sigma_v^2 / T) + \lambda^2 (\sigma_a^2 + \sigma_v^2 / T) \\ &= \sigma_a^2 - 2(1 - \sqrt{\eta} / \sqrt{\gamma}) \gamma + (1 - \sqrt{\eta} / \sqrt{\gamma})^2 \gamma = \sigma_a^2 + \eta - \gamma = 0. \end{aligned}$$